

Assignment 3

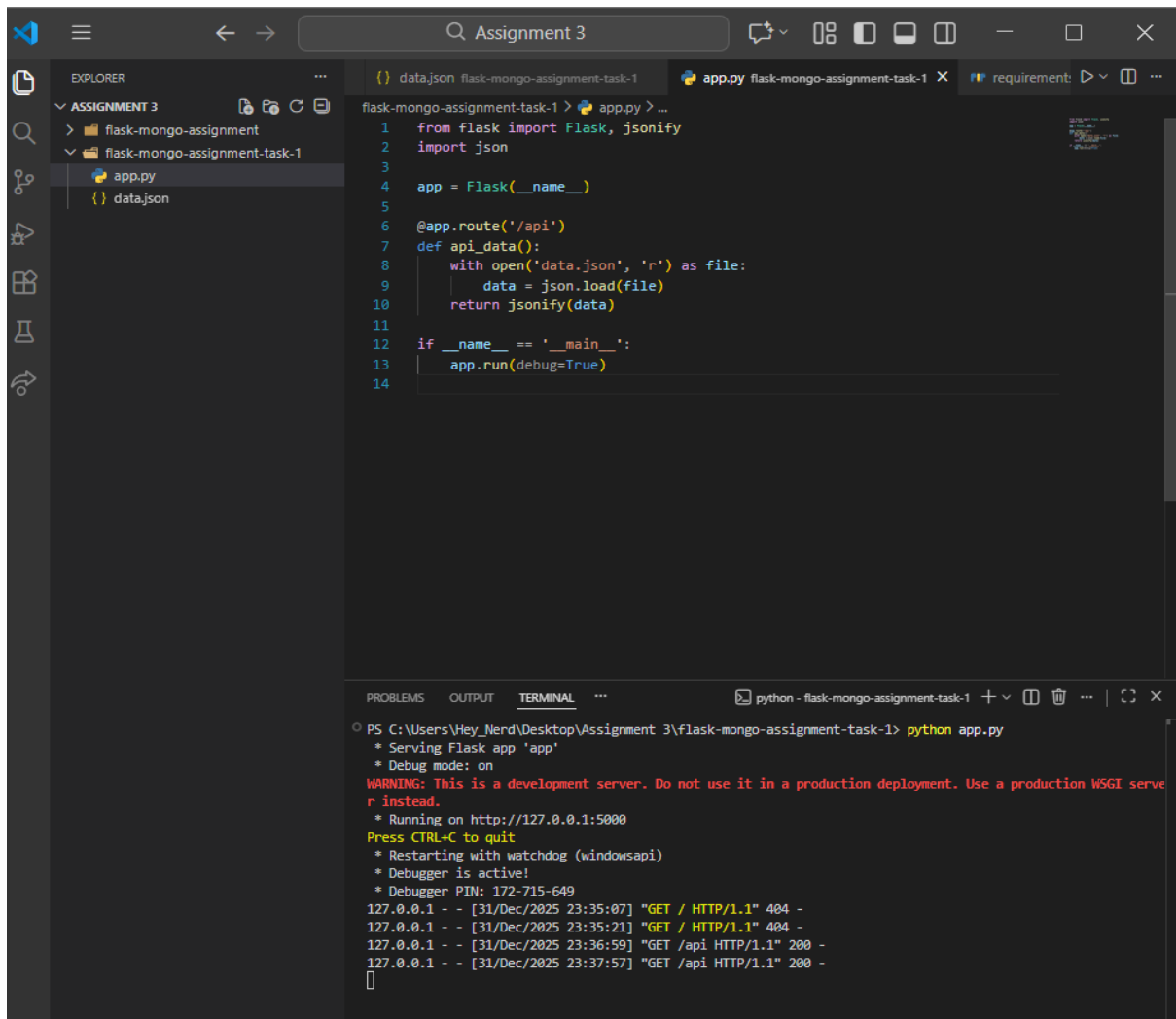
1. Create a Flask application with an /api route. When this route is accessed, it should return a JSON list. The data should be stored in a backend file, read from it, and sent as a response.

Solution:

The Flask application contains only the /api route. When accessed through /api, it successfully reads data from a backend JSON file and returns it as a JSON response.

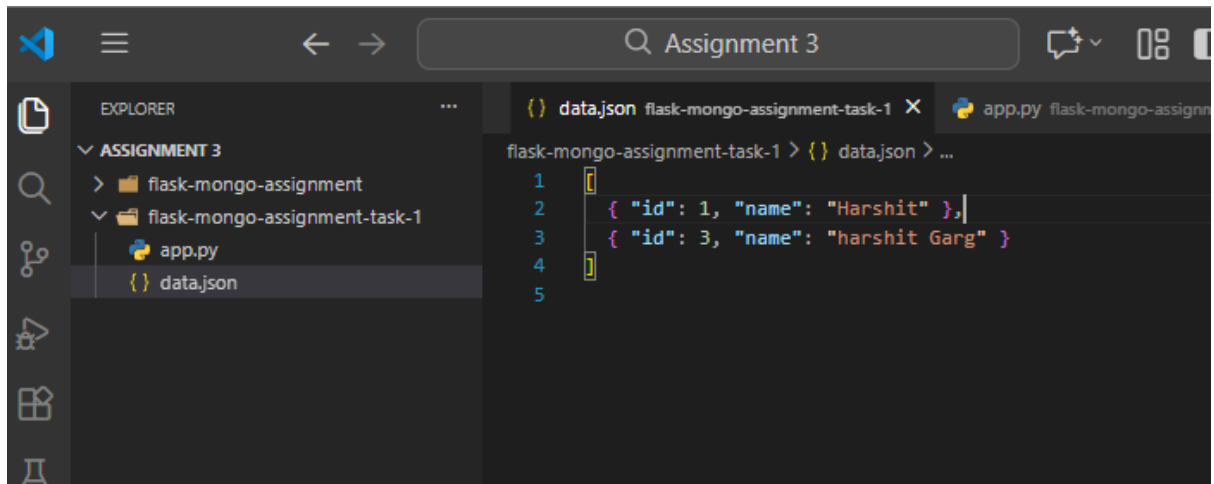
app.py

```
from flask import Flask, jsonify
import json
app = Flask(__name__)
@app.route('/api')
def api_data():
    with open('data.json', 'r') as file:
        data = json.load(file)
    return jsonify(data)
if __name__ == '__main__':
    app.run(debug=True)
```

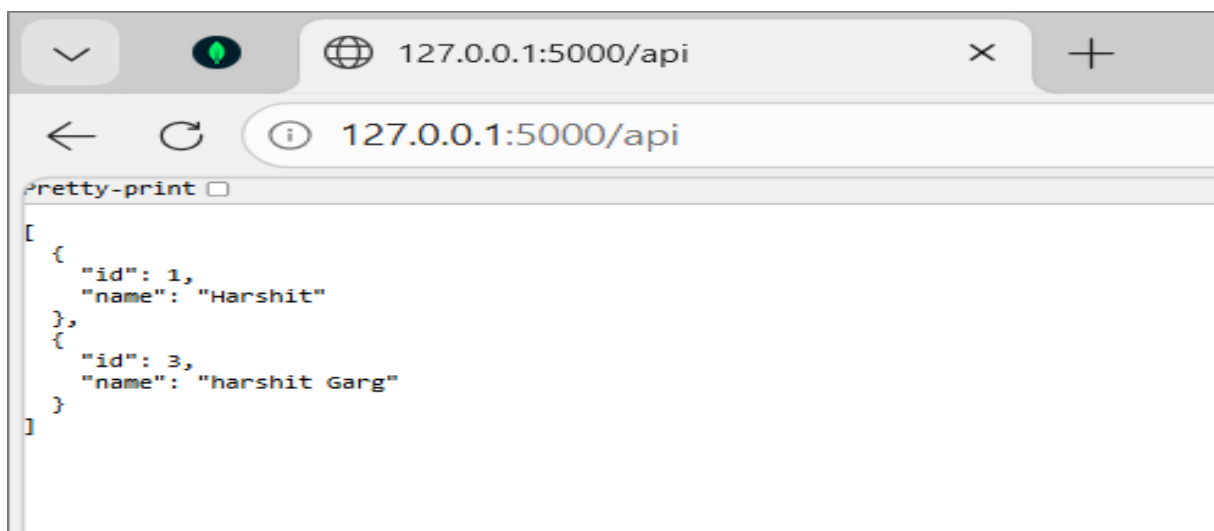


Data.json

```
[
  {"id": 1, "name": "Harshit"},
  {"id": 3, "name": "harshit Garg"}
]
```



/api route Data fetch



2. Create a form on the frontend that, when submitted, inserts data into MongoDB Atlas. Upon successful submission, the user should be redirected to another page displaying the message "Data submitted successfully". If there's an error during submission, display the error on the same page without redirection.

Solution:

A frontend HTML form was created and connected to a Flask backend. On form submission, the data is inserted into MongoDB Atlas using PyMongo; if the insertion is successful, the user is redirected to a success page displaying “Data submitted successfully”. If any error occurs during insertion, it is caught using error handling and the error message is displayed on the same page without redirection.

Templates/form.html

```
<!DOCTYPE html>

<html>

<head>

    <title>Student Form</title>

</head>

<body>

    <h2>Submit Data</h2>
```

```

{% if error %}

    <p style="color:red;">{{ error }}</p>

{% endif %}

<form method="POST">

    <input type="text" name="name" placeholder="Enter Name" required><br><br>

    <input type="email" name="email" placeholder="Enter Email" required><br><br>

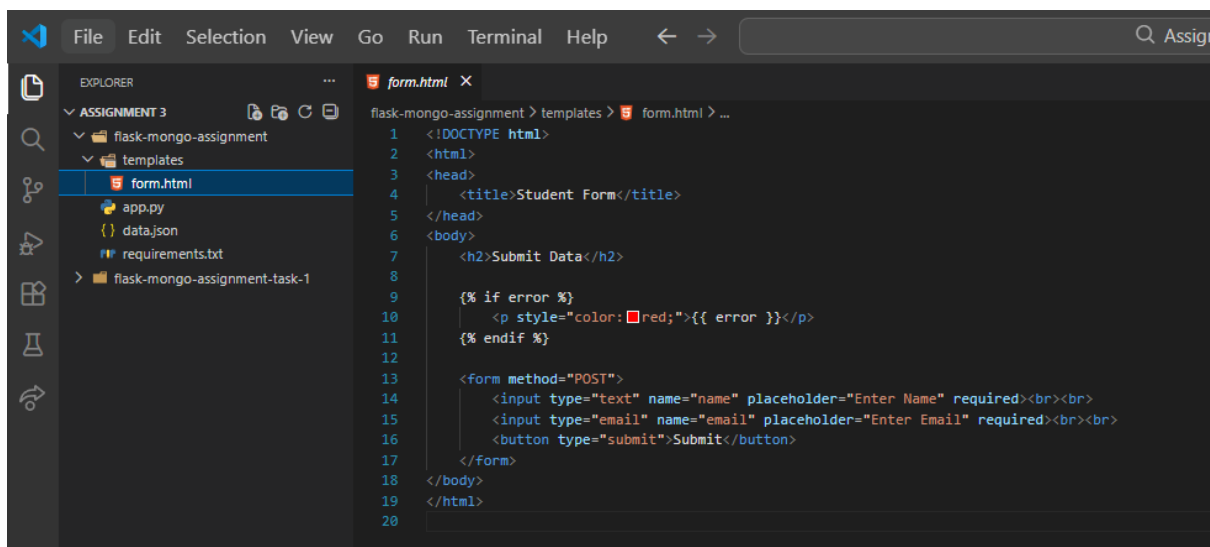
    <button type="submit">Submit</button>

</form>

</body>

</html>

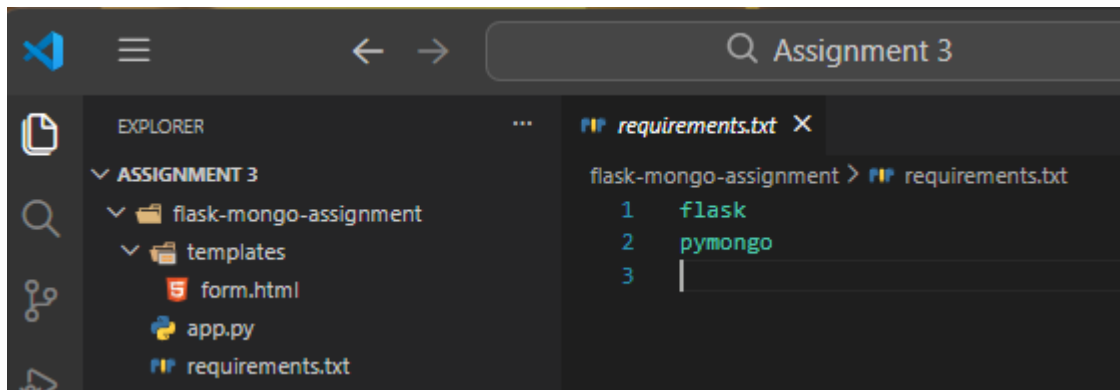
```



Requirements.txt

flask

pymongo



app.py

```
from flask import Flask, jsonify, render_template, request, redirect
```

```
from pymongo import MongoClient
```

```
import json
```

```
app = Flask(__name__)
```

```
@app.route('/api')
```

```
def api_data():
```

```
    with open('data.json', 'r') as file:
```

```
        data = json.load(file)
```

```
    return jsonify(data)
```

```
client = MongoClient("mongodb+srv://gargharshit765:<mongodbpass>
```

```
@cluster0.tohtais.mongodb.net/?appName=Cluster0")
```

```
db = client["studentDB"]
```

```
collection = db["students"]
```

```
@app.route('/', methods=['GET', 'POST'])
```

```
def form():
    error = None
    if request.method == 'POST':
        try:
            name = request.form['name']
            email = request.form['email']
            collection.insert_one({
                "name": name,
                "email": email
            })
            return redirect('/success')
        except Exception as e:
            error = str(e)

    return render_template('form.html', error=error)

@app.route('/success')
def success():
    return "Data submitted successfully"

if __name__ == '__main__':
    app.run(debug=True)
```

```
1 from flask import Flask, jsonify, render_template, request, redirect
2 from pymongo import MongoClient
3 import json
4
5 app = Flask(__name__)
6
7 @app.route('/api')
8 def api_data():
9     with open('data.json', 'r') as file:
10         data = json.load(file)
11     return jsonify(data)
12
13 client = MongoClient("mongodb+srv://gargharshit765:hello_garg765@cluster0.toht
14 db = client["studentDB"]
15 collection = db["students"]
16
17 @app.route('/', methods=['GET', 'POST'])
18 def form():
19     error = None
20     if request.method == 'POST':
21         try:
22             name = request.form['name']
23             email = request.form['email']
24             collection.insert_one({
25                 "name": name,
26                 "email": email
27             })
28             return redirect('/success')
29         except Exception as e:
30             error = str(e)
31
32     return render_template('form.html', error=error)
33
34 @app.route('/success')
35 def success():
36     return "Data submitted successfully"
37
38 if __name__ == '__main__':
39     app.run(debug=True)
40
```

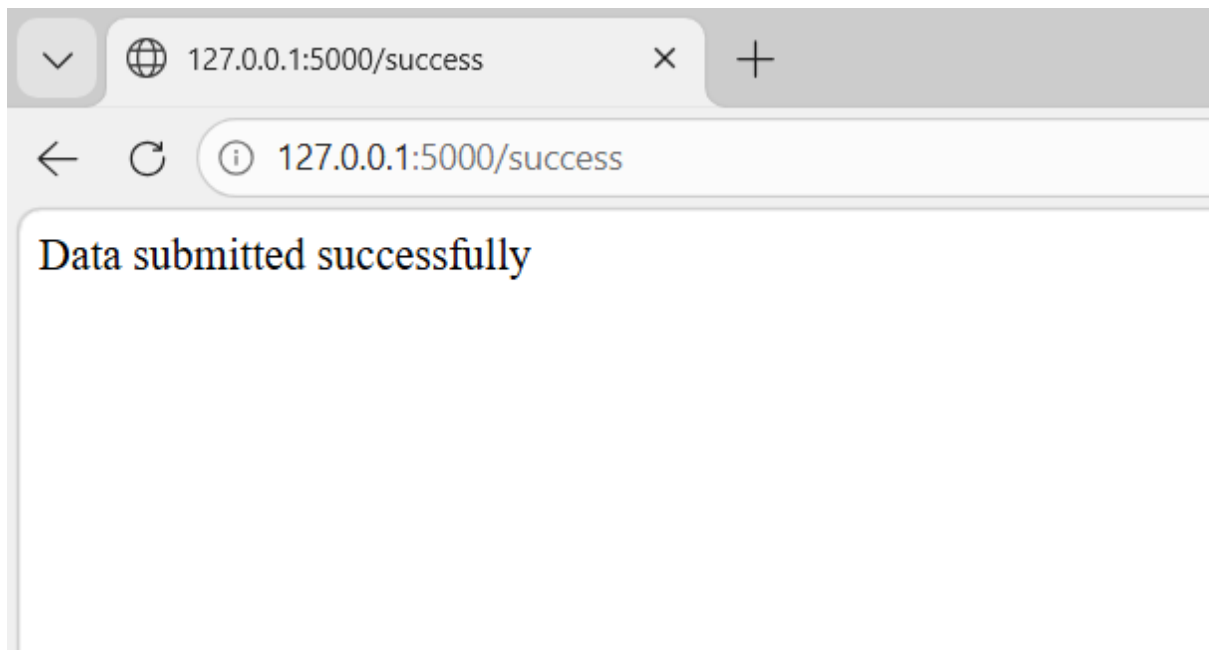
Html Form For Data Submission

Student Form

127.0.0.1:5000

Submit Data

Data Submitted Properly Message On /success route



MongoDB Atlas Page Getting Data from Html Form

